Research Background Inquiry Form

This document is a sample for your reference. Please fill in your information under the Research Background Inquiry Form (RBIF) tab in your CMS portal. Let us know if you have any questions as you complete it.

1. Please list your main academic field:

Instructions: Please fill out this prompt by listing your main academic field. Your main academic field should be the most appropriate and distinctive field under which the majority of your current and proposed research falls. In other words, your field should reflect the area in which you are making, or intend to make, significant contributions in the United States.

Some appropriate fields include: “computational modeling,” “immunology,” “applied machine learning,” “civil engineering,” “ecology,” etc.

NOTE: For NIW petitions, please ensure that the main academic field you choose aligns with your proposed endeavor. Your proposed endeavor is your “research mission statement,” which you completed earlier in the case preparation process. It specifically outlines the work that you plan to carry out in the United States. The field you choose should be the academic (or technical) category that encompasses the work outlined in your proposed endeavor. For instance, if your proposed endeavor is focused on developing AI tools for infrastructure monitoring, then your field may be described as “applied machine learning” or “computational civil engineering.” Similarly, if your proposed endeavor involves improving public health outcomes through diagnostic imaging, your field may be “cardiovascular medicine” or “biomedical engineering.” The better this alignment, the more coherent and persuasive your case will be to USCIS.

Things to Consider:

i. Please try to avoid using a field that is too broad. Choosing overly general categories can make it harder for the USCIS to understand your specific area of expertise. For example, "engineering" or "medicine" are too vague. More specific subfields such as "mechanical engineering," "civil engineering," or "cardiovascular medicine" are more effective. The key is to identify a field that reflects your actual and proposed work, while also avoiding unnecessary vagueness.

ii. Please try to avoid using a field that is too narrow. Selecting fields like "finite element modeling of composite bridge joints" is overly restrictive and may not cover the full scope of your work. Broader, but still accurate, terms like "structural engineering" or "computational mechanics" provide more flexibility.

iii. Additionally, we recommend selecting a single overarching field unless your work clearly and logically spans more than one area with a strong connection between them. For example, while "electrical and computer engineering" is acceptable, "physics and finance" is not unless there is a clear and documentable link.

iv. In many cases, the field that most accurately describes the majority of your current and proposed research projects will be closely related to your most advanced degree. For example, if you have a Ph.D. in physics, it is very likely that your main academic field will either be “physics” or some sub-area of physics. That being said, *your field does not need to be an exact match with the field of your most advanced degree.*

v. While the field you choose should align first and foremost with your current and future research, you should consider your body of work as a whole. Ideally, your chosen field should appropriately align with your past achievements, such as your past research projects, degrees, or work experiences, just as much as it aligns with your current and proposed research.

   If you plan to transition into a new field, select a field that connects both your past achievements and your proposed research. In other words, please ensure that the new field can, in some capacity, be justified by your past achievements. For example, if your Ph.D. is in physics, but your current work involves financial modeling, you may want to choose a field like “computational modeling” or “applied data,” since these are interdisciplinary fields

that make sense in “physics” and “financial modeling” contexts. Similarly, if your degree is in civil engineering, but you are currently working in a tech-forward role developing machine learning models, you may want to choose “applied machine learning,” “data science,” or “computational modeling,” since these fields involve aspects of both “engineering” and “technology/computer science.”

In sum, you should be choosing a field that specifically and accurately describes your body of work, can be supported with past achievements, publications, skills, and future endeavors, and clearly aligns with your proposed future work in the United States.

Why we ask this question: A main academic field is critical for NIW petitions. In fact, your chosen field stands at the very center of the story we are presenting to the USCIS about you and your accomplishments. Across the many types of documents that we will be submitting as part of your petition to the USCIS (recommendation letters, petition letters, employment letters, etc), your main academic field will remain a consistent reference point, allowing us to present your achievements in a coherent and specific way. It is therefore important that we select the correct and most accurate field for your case.

Infectious Diseases

2. Please list up to three CET categories, sub-areas, or specializations that your work contributes to:

Instructions: Please fill out this prompt by listing up to three CET categories (see below for more instructions), sub-areas, or specializations that your work contributes to. These CET categories, sub-areas, or specializations should fall under the scope of your main academic field. They can reflect either the nature of your work or distinct specializations that you hold and which distinguish you from your peers. Again, these CET categories, sub-areas, or specializations should reflect the majority of your current and proposed research, while also not being too distinct from your past accomplishments.

“CET categories” refer to areas of research listed in the National Science and Technology Council’s “Critical and Emerging Technologies.” For background, in 2020, the National Science and Technology Council (NSTC), through the White House Office of Science and Technology Policy (OSTP), developed a policy document that identified and assessed technologies that are critical to U.S. national security and economic competitiveness. Since then, this document has been updated on two occasions: 2022 and 2024. As it currently stands, the CET list contains 18 areas of research, each of which contain numerous sub-areas. Please look through the following document (https://www.govinfo.gov/content/pkg/CMR-PREX23-00185928/pdf/CMR-PREX23-00185928.pdf) and identify up to three areas or subareas that are pertinent to your research. If you are unable to identify appropriate CET categories, please feel free to list sub-areas or specializations that are not listed in the CET list.

Some appropriate answers to this question include: “CET category: Advanced supercomputing, including for AI applications” (main field: computer science), “CET category: Batteries” (main field: materials science), “CRISPR-Cas9 genome editing” (main field: virology), “computational fluid dynamics” (main field: computational modeling), “natural language processing” (main field: applied machine learning), “urban and regional planning” (civil engineering), etc.
NOTE: For NIW petitions, please ensure that your CET categories, sub-areas, or specializations not only fall under the scope of your main academic field, but also align with the contents of your proposed endeavor statement.

Why we ask this question: Your three CET categories, sub-areas, or specializations are important for NIW petitions. In fact, they are critical to the development of the story we are presenting about you to the USCIS. The USCIS judges thousands of immigration petitions each year. In order for us to distinguish your petition, we must rely on emphasizing your specializations and irreplaceable skills. Across the documents we submit to the USCIS, these CET categories, specializations, and sub-areas are emphasized. Additionally, they need to remain consistent in order to ensure that the USCIS values and recognizes your distinctive capabilities.

Disease Diagnosis
COVID-19
Quality Improvement in Healthcare

Instructions: To complete this section, please describe the importance of your current and/or future research in the United States. Specifically, you should be describing how your current and future research benefits: (a) your field, (b) any global issues, (c) issues related to the United States. If possible, please try to address all three impacted areas in your response to this prompt.

Although you can fill out this section of the Research Background Inquiry Form any way you see fit, you may want to organize your discussions about the importance of your research by following the prompts below:

i. **Describe the problem your research is attempting to address**
   - What key problem or challenge does your current or future research in the United States aim to solve?
   - Why is this problem urgent or important to address now?
   - What gap currently exists in the field?
   - How will your proposed work address this gap or improve upon current solutions?

ii. **Describe the current or potential impact of your research**
   - Who will benefit from your proposed work (e.g., patients, industry, government, academic community)?
   - What real-world changes might result from your work (e.g., improved health outcomes, new technologies, more efficient systems)?
   - Does your work reduce costs, improve safety, promote sustainability, or enhance national security?

iii. **Describe the broader relevance your work will have on your field, global issues, or the United States**
   - How will your work benefit your field?
   - Is your work aligned with any global priorities (e.g., climate change, public health, security, the economy, etc.)?
   - Is your work aligned with any U.S. government priorities (e.g., artificial intelligence, clean energy, infrastructure, pandemic preparedness, national security)?

Here are some examples of appropriate responses to this section of the Research Background Inquiry Form:

Example #1:

National Issues: *My research focuses on using AI models to improve early detection of cardiovascular disease in underserved populations—a pressing public health challenge in the United States. Cardiovascular disease is the leading cause of death across the country, responsible for about 1 in every 5 deaths. The burden is particularly high in rural and low-income communities, where access to regular screening and diagnostics is limited. By developing scalable, data-driven AI tools that analyze electronic health records and wearable sensor data, I aim to improve the identification of high-risk individuals in these communities—enabling earlier, more accurate intervention.*

The national importance of this endeavor lies in its alignment with multiple U.S. priorities. It supports the Department of Health and Human Services' digital health transformation agenda and complements the goals of the Biden Administration's Cancer Moonshot and health equity initiatives. More broadly, it contributes to AI innovation in healthcare—a domain identified as a critical and emerging technology by the National Science and Technology Council. Ultimately, this work could enhance public health outcomes, reduce long-term medical costs, and contribute to a more equitable healthcare system in the United States.

Global Issues: *This work also has global relevance because it addresses a preventable condition with enormous societal and economic costs. Without better tools for early detection, many individuals will continue to be*

diagnosed only after severe or irreversible complications occur, resulting in higher mortality, lower quality of life, and increased healthcare expenditures.

Example #2:

My proposed endeavor involves creating data-driven decision support systems that help U.S. cities prepare for and respond to the growing threat of climate change. Specifically, I will design predictive tools using advanced analytics to assess vulnerabilities in infrastructure and guide policy and investment decisions for resilience planning. Climate-related disasters such as floods, hurricanes, and heatwaves are becoming more frequent and severe, posing significant risks to public safety, critical infrastructure, and local economies. Without forward-looking tools that support evidence-based planning, many communities will continue to suffer from preventable damage and costly recovery processes.

This work holds substantial merit because it addresses a key national challenge—enhancing climate resilience through technology. The U.S. Government Accountability Office has consistently identified climate risks to infrastructure as a top fiscal exposure for the country. My decision support systems aim to mitigate these risks by enabling municipalities to invest more efficiently, reduce future disaster losses, and safeguard communities. This endeavor contributes to U.S. national interests by aligning with major federal initiatives such as the Bipartisan Infrastructure Law, FEMA's climate resilience programs, and the National Climate Adaptation Strategy. Ultimately, my tools will empower local governments with actionable insights, promote smarter public spending, and support long-term environmental and economic sustainability.

Again, you are welcome to fill out this section of the Research Background Inquiry Form in whatever way feels right to you. You may use freeform paragraphs, or you may follow the prompts provided to guide your response in a structured way. If you have questions about how to frame this section or which format might be more effective for your case, feel free to message us.

Things to Consider:

i. Please be specific when completing this section. Additionally, please use plain language that a non-expert can understand.

ii. Please feel free to cite external reports, U.S. government priorities, national strategies, or publicly available sources to support the importance of your research, the problems you are seeking to address, and your future solutions to these problems. If you reference webpages, please include the link in your answer.

iii. While it is acceptable to reference your past research or achievements to support the value and feasibility of your proposed endeavor, this section must emphasize the work you are, or will be doing, in the United States.

Why we ask this question: For NIW petitions, the information provided in this section supports the "substantial merit and national importance" (Prong 1) prong under the EB-2 NIW standard (Dhanasar). In several of the documents we are drafting for your case (recommendation letters, petition letters, Proposed Endeavor Statements, etc), there will be extensive discussions about the benefits, importance, and applications of the research you are, or will be, conducting in the United States. These arguments are absolutely critical to your petition to the USCIS. In fact, these arguments are the most consequential facet of the story we are presenting about you to the USCIS. They verify the indisputable need for the United States to accept you into the country and contribute to its central industries.

My research addresses major global health challenges, with a particular focus on infectious diseases such as COVID-19, malaria, and childhood diarrheal illnesses. At the onset of the COVID-19 pandemic, when high-quality evidence on clinical presentation, laboratory abnormalities, and imaging features was limited, I led a comprehensive systematic review that synthesized the best available studies to guide prompt diagnosis, clinical management, and prognostication. This work not only supported frontline decision-making, but also strengthened the evidence base for early detection and public health response during emerging outbreaks. Building on this foundation, I conducted comparative global analyses of COVID-19 transmission control strategies across multiple international cities and provinces. These studies introduced the population-level "Swiss cheese model" to explain how layered interventions interact offering policymakers a practical, lower-disruption alternative to large-scale lockdowns.

Beyond COVID-19, I have contributed to strengthening infectious disease control in other high-burden areas. My systematic review of antimalarial drug development synthesized high-quality evidence on therapeutic targets and emerging drug-resistance patterns, helping inform future treatment innovation. Additionally, my investigation of diarrheal pathogens among children under five identified the leading infectious agents driving dehydration and morbidity in resource-limited settings, providing essential evidence for targeted prevention and treatment programs.

Collectively, this body of work reflects my commitment to generating rigorous, policy-relevant evidence that improves disease prevention, guides clinical and public health decision-making, and enhances national and global preparedness for future outbreaks.

4. Please describe up to three papers/publications that reflect the progress you have made towards your proposed endeavor:

Instructions: Please describe three of your publications that best represent your accomplishments and the progress you have made in your field. If possible, we recommend that you select your **top-cited publications**. That said, we can also proceed with your other impactful publications, such as those where you are the first or lead author on a publication, those published in prestigious journals, or those supported by significant funding.

Please also ensure that the publications you choose **align with your proposed endeavor**.

This section contains three distinct prompts.

- After the **first prompt** ("In:") please provide the title and any other bibliographic information (journal name, year of publication) of the publication you would like to discuss.

- After the **second prompt** ("I described:"), please provide a 100 word description of your publication. Although you can fill out this section however you see fit, to help you organize your thoughts, you can follow the guide questions below:
    i. Describe the problem your project set out to explore and solve
    ii. Describe the steps (methodology) you took to explore or solve that problem
    iii. Describe the outcome (results) of your actions

- After the **third prompt** ("The research described in this paper advances my proposed endeavor by"), please provide a 50-100 word statement about how your paper aligns with your proposed endeavor. This section should make it clear that this publication represents a tangible step towards making progress on your proposed endeavor.

Here is an example of an appropriate response to this section of the Research Background Inquiry Form:

In: "Low temperature reaction kinetics inside an extended Laval nozzle: REMPI characterization and detection by broadband rotational spectroscopy" published in the *Journal of Chemical Physics* (2023)

I described: I led the development of a cutting-edge experimental system that combines an extended Laval nozzle with CP-FTmmW and REMPI spectroscopy to study low-temperature reaction kinetics under uniform supersonic flow. This setup enables precise, quantum-state-specific detection of radicals and molecules in dense, cold environments, overcoming technical challenges such as FID signal disruption due to high collision frequencies. I specifically designed and validated a Laval nozzle capable of generating a uniform 20 K flow, with a shock-free secondary expansion reaching temperatures as low as 3.9 K. CFD simulations and REMPI confirmed flow uniformity, eliminating the need for invasive impact pressure measurements. This work produced the first measurement of the HCO + O_2 reaction rate at 20 K, yielding critical data for modeling chemical processes in interstellar and planetary atmospheres. My contributions demonstrate technical innovation and leadership in advancing astrochemical and atmospheric science, aligning with national priorities in climate, space exploration, and sustainable energy.

The research described in this paper advances my proposed endeavor by: establishing a novel experimental framework for studying low-temperature reaction kinetics and allows for direct investigation of chemical dynamics in extreme environments, aligning closely with my goal to improve predictive models for atmospheric, combustion, and astrochemical processes. One of the key outcomes, namely the first-ever measurement of the HCO + O_2 reaction rate at 20 K, provides essential kinetic data for modeling interstellar and planetary chemistry. The use of computational fluid dynamics (CFD) to validate flow conditions further ensures experimental accuracy. This work therefore lays a critical foundation for advancing our understanding of chemical reactivity in cold environments, reinforcing the central objectives of my proposed scientific endeavor.

Why we ask this question: Your primary publications are a crucial component of your petition. The selected publications will be discussed in detail in your petition letter and support letters (if applicable), which play a central role in presenting your story to the USCIS for them to verify the tangible and unique skills that you have. They are also proof to the USCIS that you are capable of high-impact, impressive achievements that the United States would benefit from having.

A. In

"Clinical and laboratory factors associated with coronavirus disease 2019: A systematic review and meta-analysis"

I described

the most up-to-date COVID-19 data and made suggestions to address some of the disease's major clinical roadblocks. At the onset of the COVID-19 pandemic, the global scientific community lacked high-quality evidence on the disease's clinical presentation, laboratory markers, and imaging features. To fill this gap, we conducted a rigorous systematic review of all available high-quality studies, synthesizing data from diverse international cohorts. Our analysis identified the most consistent clinical symptoms, key laboratory predictors of severity, and characteristic radiologic patterns of COVID-19. By translating these findings into clear, evidence-based recommendations, we provided clinicians with the early guidance needed for accurate diagnosis, effective management, and timely risk stratification during a period of profound uncertainty.

The research described in this paper advances my proposed endeavor by

strengthening the evidence base that guides clinical decision-making and public health response during infectious disease outbreaks. This work enhanced my ability to translate research findings into practical clinical and preventive strategies. It also reinforced my expertise in evidence synthesis, epidemiologic evaluation, and global health preparedness, skills that are directly applicable to my ongoing efforts to improve infectious disease control and outbreak response at both national and international levels.

B. In

"How cities and provinces around the globe tackled Covid-19 outbreaks in 2021"

I described

comparative global perspectives on the effectiveness of transmission control strategies during the COVID-19 pandemic. By analyzing public health interventions across ten major international cities, I identified the key factors influencing the success and failure of outbreak control measures. Through the introduction of the population-level "Swiss cheese model," my study offered a conceptual framework to explain how overlapping preventive barriers interact, and sometimes fail, under real-world conditions. This model emphasized that pandemic control requires continuous monitoring, adaptive policy enforcement, and the reinforcement of weak protective layers, rather than reliance on any single intervention. Our findings advanced the understanding of how localized public health responses shape national outcomes, providing policymakers and health systems with insights to strengthen their response to future disease outbreaks.

The research described in this paper advances my proposed endeavor by

enhancing my ability to evaluate and design evidence-based public health interventions for infectious disease control. Through comparative analysis of COVID-19 transmission strategies across multiple global cities, this study deepened my understanding of how policy implementation, social behavior, and epidemiologic factors interact to determine outbreak outcomes. It also strengthened my expertise in data synthesis, global health policy evaluation, and modeling of transmission dynamics, all of which are central to my ongoing efforts to improve pandemic preparedness and infectious disease management at both national and international levels.

C. In

"Burden of Diarrheal Diseases Among Children Under Five in Low-Resource Settings"

I described

the infectious agents responsible for childhood diarrheal illness in resource-limited environments. By analyzing high-quality studies and surveillance data across multiple regions, I identified the predominant viral, bacterial, and parasitic pathogens driving dehydration, hospitalization, and early childhood morbidity. My work highlighted not only the regional variability in pathogen prevalence, but also the environmental, nutritional, and socioeconomic factors that amplify risk. Through this evidence synthesis, we clarified where existing prevention tools, such as rotavirus vaccination and point-of-care diagnostics, are most impactful, and where critical gaps remain. These findings offered public health programs a clearer roadmap for designing targeted intervention protocols.

The research described in this paper advances my proposed endeavor by

strengthening my ability to develop and evaluate evidence-based strategies to reduce infectious disease morbidity in high-risk populations. It also enhanced my capacity to translate complex pathogen-specific findings into actionable policy recommendations, skills that are central to my broader goal of improving health equity, optimizing resource allocation, and supporting national and international efforts to protect vulnerable populations from preventable infectious diseases.

5. Please describe up to three examples of how other researchers have used your methods and findings:

Instructions: Please fill out this section by describing three examples of other researchers citing your work in their papers. Although you may answer this question however you choose, please try to ensure that each description contains bibliographic information about the paper that cited your work, 30-50 words about the topic of the paper, and 30-75 words about how the paper utilized your work. Please also make sure to provide one example for every paper you listed in Section 4.

Please note that these citations should be from independent researchers (i.e., researchers whom you have not co-authored a paper with), and from papers published in peer reviewed publications.

NOTE: If you are unable to provide three examples of other researchers citing your work, you can discuss other types of implementation. Examples of such implementation include:

- References in reviews or meta-analyses
- Adoption by curricula
- Adoption by industry
- Software or tools that implement/adapt your research
- Inclusion in government reports, white papers, or consultancies
- Use in product development
- Use by NGOs or advocacy groups
- Citations in a patent

If you have no tangible implementations at all, you may answer this prompt by describing the significance of your paper in your academic field, on issues of global significance, or on issues related to the United States.

An appropriate answer to this prompt includes:

1. In their paper titled "Twins in rotational spectroscopy: Does a rotational spectrum uniquely identify a molecule?" published in The Journal of Chemical Physics in 2024 authors Marcus Schwarting et al. cited my paper "Low temperature reaction kinetics inside an extended Laval nozzle: REMPI characterization and detection by broadband rotational spectroscopy". This paper references my work while discussing the capabilities of high-resolution rotational spectroscopy, particularly in distinguishing molecular isomers, conformers, and vibrational states. The authors emphasize that broadband chirped-pulse Fourier transform microwave (CP-FTMW) spectroscopy, the method that we use, has revolutionized the study of chemical reaction dynamics and kinetics by enabling rapid acquisition of spectral data with high resolution. They acknowledge my research in the context of using CP-FTMW spectroscopy for identifying reaction intermediates with structural and quantum state selectivity, highlighting its significance in advancing gas-phase spectroscopy and dynamics.

Why we ask this question: We like to ask this question because it is valuable to the USCIS. In fact, your answers help demonstrate to them the significance, originality, and impact of your research within your field and beyond. By asking you to describe how your work has been cited or implemented, particularly by independent researchers, we can present to the USCIS objective evidence of your national or international recognition, how your work has had tangible effects within and beyond academic publications, and show that your influence extends beyond close collaborators.

A.

Thompson, et al. "Clinical and Laboratory Factors Associated with Coronavirus Disease 2019: A Systematic Review and Meta-analysis." *Journal of Global Infectious Medicine*, 2021. Thompson et al. cited "Clinical and Laboratory Factors Associated with Coronavirus Disease 2019: A Systematic Review and Meta-analysis" as a foundational source for their discussion of early diagnostic challenges in COVID-19. They used my paper's synthesis of the most up-to-date clinical, laboratory, and imaging data to justify their selection of core severity markers and to support the development of their revised clinical risk-stratification model. The authors specifically referenced the review's identification of consistent symptoms and predictive laboratory abnormalities as evidence that standardized diagnostic thresholds were both feasible and necessary. By drawing on these findings, Thompson et al. strengthened the rationale for their proposed clinical pathway and demonstrated that their recommendations were aligned with the highest-quality evidence available during the early phase of the pandemic.

B.

Martinez et al. "Adaptive Public Health Governance During Global Crises: Lessons from COVID-19 Policy Failures and Successes." *International Journal of Pandemic Policy*, 2023. Martinez et al. cited "How Cities and Provinces Around the Globe Tackled COVID-19 Outbreaks in 2021" to underpin their argument that effective pandemic management requires coordinated, multi-layered interventions rather than isolated policy measures. They used my comparative analysis of ten international cities to illustrate how differences in enforcement, timing, and community adherence influenced outbreak trajectories. The authors specifically relied on my "Swiss cheese model" framework to explain the cumulative value and weaknesses of stacked preventive strategies. Incorporating these insights allowed Martinez et al. to strengthen their central claim that adaptive, continuously monitored public health systems are essential for minimizing policy failures in future global crises.

C.

Nguyen and Mensah. "Targeted Interventions for Childhood Diarrheal Diseases: Evidence-Based Approaches in Low- and Middle-Income Countries. *Global Child Health Review*, 2022. Nguyen and Mensah cited "Burden of Diarrheal Diseases Among Children Under Five in Low-Resource Settings" to support their recommendations for prioritizing region-specific interventions. They used my study's synthesis of viral, bacterial, and parasitic pathogens to identify which pathogens contributed most to morbidity and hospitalization in different regions, and drew on my analysis of environmental, nutritional, and socioeconomic risk factors to guide resource allocation. By integrating these findings, Nguyen and Mensah strengthened their argument for implementing targeted prevention measures.

Supplementary Information

1. Please list any funding you or your papers have received from notable U.S.-based organizations:

Instructions: Please fill out this prompt by listing all the notable, U.S.-based funding you, your papers, or your research projects have received. Ideally, this funding would only include prestigious or competitive external funding, not internal funding from the university or employer you are affiliated with. If possible, include the name of the funding body and/or program, the grant number for the funding, and the PI name. Please also indicate whether or not that funding is related to any of the publications you described above.

Some appropriate answers include: "National Institutes of Health (R01CA234567)," "National Science Foundation Award #2041234 (PI: Dr. F.A. Smith)," etc.

Department of Energy (National Nuclear Security Administration) under Award Number DE-NA0004104 and Contract number B647887 (PI: Dr. Roger H. French)' , 'U.S. National Science Foundation Award under Award Number 2133576 (PI: Dr. Roger H. French)'

This funding is associated with the following papers in my Research Background Inquiry Form: 1

2. Please list any nationally or internationally recognized prizes or awards you have received for your research:

Instructions: Please fill out this prompt by listing any nationally or internationally recognized prizes or awards that you have received. Although you can fill out this prompt however you choose, ideally, the awards listed here would only include prestigious or competitive external awards, not internal awards granted by your university or employer.

Award in Infectious Disease Research, American Society of Tropical Medicine and Hygiene (ASTMH), 2021: Awarded for research advancing knowledge of malaria therapeutics and pathogen-specific interventions in low-resource settings.

3. Please list any instances where your work has received media coverage or been written about in newspapers, magazines, or other widely circulated media:

Instructions: Please list all instances where you or your research have been the subject of media coverage. This may include newspaper or magazine articles, profiles in professional or trade publications, TV or radio interviews, or online coverage by credible outlets. Articles that are primarily about your research, impact, or achievements are most relevant here, as well as articles that explicitly state your name.

When answering this question, please avoid mentioning marketing material, press releases from your employer or affiliated university, coverage in college newspapers, or any reports that do not explicitly include your name.

N/A

4. Please describe any occasions where you have been invited to peer review the work of other researchers, and if you are on any editorial boards:

Instructions: Please fill out this prompt by listing all the journals or conferences for which you have been invited to conduct peer review. Peer review is the evaluation of scholarly work by experts in the same field to ensure its quality, validity, and relevance before publication or acceptance. When listing your peer review experience, please clearly write the journal or conference name and the number of reviews you have conducted for that specific journal or conference. Additionally, if you are on any editorial boards, you may list them here as well.

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5. Please describe any instances where your research has been commercialized, licensed, or resulted in the creation of new products, patents, or startups:

Instructions: Please describe any ways in which your research has been used for commercial applications. This may include technology transfer, licensing agreements, patents that have been adopted in industry, spin-off or startup companies you have founded or contributed to, products and services that were developed directly from your work, or widely used tools and software. You may also include collaborations with industry that led to tangible commercialization outcomes.

N/A

6. Please list any leading or critical roles you have held at well-regarded companies, institutions, or organizations:

Instructions: Please describe any leadership or critical roles you have held in well-regarded organizations, companies, institutions, or teams. Include roles where your contributions were essential to the success of major projects, and provide context about the reputation or prestige of the organization itself. Your answer may include executive or senior management positions in leading companies, research leadership positions, principal or named investigator positions for departments, institutions, or businesses that receive merit-based government awards, creative director or lead design roles, among others.

N/A

7. Please list any open-source contributions:

Instructions: Please describe any open source projects to which you have contributed. When filling out this information, please provide the following:

1. *Project name and relevant links:* Please share the names of the open-source repositories or projects that you have contributed to, along with public links (ex. GitHub, GitLab, etc).

2. *Role:* Please briefly describe the nature of your contribution. For example, you may describe the specific modules, features, or documentation you developed, any bugs you fixed, and/or any code reviews you completed.

3. *Impact and Adoption:* Please provide any relevant metrics, if possible. These metrics can include the number of downloads or forks, stars or followers on the repository, references to your code in other projects or technical documents, or any mention of your work in industry articles, blogs, or conference talks.

4. *Connection to U.S. priorities:* If applicable, please explain how your open-source work contributes to areas of national interest, such as clean energy, cybersecurity, advanced manufacturing, or artificial intelligence/machine learning technologies.

5. *Recognition Documents:* If you have received recognition for your contributions or developments, please provide those as well. Types of received recognition may include thanks in release notes, community awards, invitations to maintain projects, etc.

An appropriate answer to this prompt may read:

Project 1: GreenGridAI (github.com/greengrid/GreenGridAI)
Role: I designed and implemented the forecasting engine module (Python) and improved prediction accuracy (MAE reduced by ~15%). I also added documentation, usage examples, and integrated CI/CD with GitHub Actions. Additionally, I mentored 5 first-time contributors via code reviews and issue triage.
Impact & Adoption: 1,250 stars, 80 forks. Downloaded ~10,000 times via pip. Integrated into two community solar-energy dashboards.
U.S. Priorities: This project enhances clean energy forecasting—crucial for renewable grid optimization and sustainability goals.
Recognition: Listed in release notes as “Key contributor implementing forecasting enhancements.”

N/A

8. Please describe any additional accomplishments that you feel are important, but have not been covered in this Research Background Inquiry Form:

Instructions: You can fill out this prompt however you choose. If you feel as though any of your primary contributions or achievements have not been covered in the Research Background Inquiry Form above, please describe them here. We will pay special attention to any information you provide in this section.

Although you may fill out this section however you wish, we have provided some claimable accomplishments and questions to help guide your thoughts:

- Notable Memberships: Have you been, or are you, a member of any selective or prestigious professional organizations? (Note: In our experience, we have found the USCIS particularly values high-ranking memberships such as “senior member” or “fellow”. However, this should not prevent you from listing other membership types in this section.)

- High Salary: Are you earning in the top 25% of your field or have any notable compensation packages?

Senior Member, International Society for Infectious Diseases (ISID): Elected based on demonstrated leadership and impact in global outbreak response, including COVID-19 research and public health intervention development.